

Year 10 Learning Checkpoint 1

Sections A-C - written theory mark scheme

Teacher copy. Keep secure. This revised written assessment totals 20 marks. Section D remains a separate 20-mark practical assessment.

Questions 01.1-01.2 reproduce the June 2025 OxfordAQA questions. Questions 01.3-03.3 are newly written school assessment questions aligned to taught 9210 content. They are not official past-paper questions.

Source and marks record

Question	Source / assessed content	Marks
01.1-01.2	June 2025 9210/2 Q02.1-02.2; QP p.3; MS p.6. Only question numbering changes.	3
01.3-01.4	School-written: applied decomposition; algorithm and implementation. Specification 3.1.1.	3
02.1-02.3	School-written: data-type choice, constants, meaningful identifiers and maintenance. Specification 3.2.1-3.2.2.	6
03.1-03.3	School-written: assignment, tracing, comparison, AND/OR and selection. Specification 3.1.1, 3.2.2, 3.2.4-3.2.5, 3.2.7.	8
Total	Written theory	20

General marking guidance

Award whole marks for the points stated. Accept equivalent wording where the meaning is clear. Do not award the same idea twice. A correct answer with an incorrect or contradictory explanation does not earn an explanation mark.

For school-written questions, minor spelling and punctuation errors do not lose marks unless they change the meaning. For pseudocode, accept clear, unambiguous alternatives, including correctly structured Python. Do not require a particular indentation width or case.

Abbreviations: **A.** accept; **R.** reject; **I.** ignore; **NE.** not enough for credit. Official guidance for 01.1-01.2 is reproduced on the next page.

Archive review excluded June 2023 Q02 and June 2024 Q05 because lesson resources already assign them as practice. It also excluded questions requiring untaught iteration, arrays, databases or logic circuits.

Section A - 6 marks

Official marking guidance: June 2025 9210/2 Q02.1-02.2

Question	Part	Marking guidance	Total marks
02	1	The process of removing unnecessary details <u>from a problem</u> ; A. "information" for "details" R. "data" for "details"	1 AO1=1

Question	Part	Marking guidance	Total marks
02	2	1 mark: Breaking a problem down (into sub problems) A. "task" for "problem" A. "program", "algorithm" for "problem" as BOD 1 mark: So that each sub problem accomplishes an identifiable task // each sub-problem might be further subdivided	2 AO1=2

Original 02.1 maps to classroom 01.1 (1 mark); original 02.2 maps to classroom 01.2 (2 marks). Source: AQA-g210-2-Final-MS-Jun25-v1.0, p.6.

01.3 Marking guidance [2 marks]

Award 1 mark for each of two distinct sub-problems described in terms of its task, for example:

- obtain/store the student identifier and quantity requested;
- compare the quantity requested with kits available to decide whether the booking can proceed;
- display the appropriate confirmation/refusal to the student.

A. other sensible subdivisions of the stated problem. **NE.** isolated labels such as "input" or "processing" without saying what they do. Do not credit the same sub-problem twice.

01.4 Marking guidance [1 mark]

An algorithm describes steps for solving a problem, whereas a program implements those steps in a programming language; or an algorithm does not have to be executable code, whereas a program is code intended for a computer to execute. Award 1 mark for a valid distinction.

R. an algorithm must be written in Python; a program and an algorithm are always the same thing.

Section B - 6 marks

02.1 Marking guidance [2 marks]

1 mark: **string**.

1 mark: explains that it preserves the identifier exactly, including the leading zeros, or that it represents a label/text rather than a quantity used in arithmetic.

A. str. **NE.** "it is a number" or "it contains digits" as a justification for string. A contradictory justification earns no reason mark. The two points are independent, but both must describe a suitable representation.

02.2 Marking guidance [2 marks]

Award 1 mark for each of these two distinct explained advantages:

- a descriptive constant name makes the purpose of the value easier to understand;
- the maximum can be updated in one definition, and every use refers to that updated value, avoiding missed or inconsistent updates.

Treat changing in one place and avoiding inconsistent updates as one maintenance point, not two separate marks.

NE. "better", "easier" or "faster" without explaining how. **R.** uppercase names automatically prevent reassignment in Python. The question assesses the purpose of a named constant, not a Python enforcement mechanism.

02.3 Marking guidance [2 marks]

1 mark: a meaningful, valid identifier such as **kits_requested**, **number_of_kits** or **requested_quantity**.

1 mark: explains that the name tells the other programmer what the stored value represents, so they can understand or change the relevant calculation/decision without guessing what n means.

A. other naming conventions with a clear meaning. **NE.** "it is meaningful" or "it is better" alone. The explanation must apply to the suggested name.

Section B total: 6 marks

Section C - 8 marks

03.1 Marking guidance [3 marks]

1 mark: Available records **12 then 10** in that order.

1 mark: Requested is **5**, Approved is **Y**, and Available subsequently becomes **5**.

1 mark: outputs **Booked then 5** in that order, with no additional output.

I. unnecessary repeats of unchanged values; blank cells; quotation marks around text; row choice, provided the sequence of events remains clear. An incorrect additional value in an assessed variable sequence loses that point. Mark each point independently.

Available	Requested	Approved	OUTPUT
12			
10			
	5		
		Y	
5			
			Booked
			5

03.2 Marking guidance [2 marks]

1 mark: AND requires both enough kits and teacher approval for the booking to proceed.

1 mark: OR would allow a booking when only one requirement holds, such as enough kits but no approval, or approval but too few kits.

A. an equivalent correct counterexample with values. **NE.** only reciting "AND means both; OR means either" without relating the rule to the booking conditions.

Section C - continued

03.3 Marking guidance [3 marks]

Award 1 mark for each requirement implemented correctly:

1. Zero and negative quantities produce **Invalid quantity**.
2. Values 1, 2 and 3 produce **Collect from office**.
3. Values greater than 3 produce **Ask technician**, and the selection produces exactly one appropriate message for every integer input.

For points 1 and 2, credit the correctly implemented path even if another path is incorrect. Point 3 requires the complete mutually exclusive decision. Do not award a further mark for the supplied input statement.

A. equivalent IF/ELSE IF/ELSE, nested IF or separate non-overlapping conditions. **I.** minor message punctuation/case differences and harmless minor pseudocode syntax errors where the intended logic is unambiguous. **R.** incorrect inclusion/exclusion of boundary values.

Example full-credit solution

```
Requested ← USERINPUT
IF Requested ≤ 0 THEN
    OUTPUT "Invalid quantity"
ELSE IF Requested ≤ 3 THEN
    OUTPUT "Collect from office"
ELSE
    OUTPUT "Ask technician"
ENDIF
```

Boundary check for this solution

Input	Expected output
-1 or 0	Invalid quantity
1, 2 or 3	Collect from office
4 or greater	Ask technician

Written total: 6 + 6 + 8 = 20 marks.

Section D remains 20 marks. Whole assessment: 40 marks.

Official questions retain their official guidance. School-written marking points and assessment judgments are local to this checkpoint; they are not an official OxfordAQA mark scheme.